

Artificial Intelligence and the Economy: Productivity, Macroeconomic Adjustment, and Policy Challenges

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1 Literature Review

Artificial intelligence differs from earlier automation technologies in what it automates. It reaches into cognitive work. Earlier technologies automated physical labor. It analyzes data. It generates text, images, and code. It can perform tasks that require creativity and judgment. It answers questions that, until recently, only trained professionals handled.

The speed and scope of adoption make the economic stakes unusually high. Organizational AI use jumped from 55 to 78% in a single year. Private investment in generative AI reached \$33.9 billion in 2024 (Stanford Institute for Human-Centered Artificial Intelligence, 2025). Roughly 80% of U.S. workers hold jobs where large language models reach at least some of their tasks (Eloundou, Manning, Mishkin, & Rock, 2024). The question is no longer whether AI will reshape the economy. It is how far that reshaping will go.

Economists have responded with a rapidly growing body of work. At the micro level, randomized experiments and workplace studies measure what happens when individual workers gain access to AI tools. At the macro level, models ask whether those individual gains translate into faster aggregate growth, and if so, by how much. The policy literature asks what governments should do if the gains prove real but unevenly distributed. The micro evidence yields consistent findings. The macro evidence does not. The order-of-magnitude disagreement between the leading macro estimates is the central tension this review examines. Resolving it requires answering a question that cuts across all three layers: how quickly do firms adopt AI and reorganize around it? Existing data captures this process poorly (Brynjolfsson, Korinek, & Agrawal, 2025). That gap runs through every section that follows.

1.1 Micro productivity and labor-market exposure

Randomized experiments measure AI's effect on worker productivity directly. Noy and Zhang (2023) run a preregistered experiment with 453 professionals. ChatGPT access reduces writing-task completion time by 40% and raises output quality by 18%. The largest gains

go to lower-performing workers. Noy and Zhang randomize assignment and blind-evaluate output. External validity is limited. The sample draws from one online platform and one task type. The authors have not tested whether these effects persist in routine work or scale to other cognitive domains. A single task on a single platform tells us something, but not everything. Whether the 40% time reduction generalizes to other forms of writing, or to other professions, is unknown.

Workplace data address some of these external validity concerns. Brynjolfsson, Li, and Raymond (2025) study 5,172 customer support agents at a Fortune 500 firm. Average productivity rises by 14%. The bottom skill quintile improves by 34%. Top-quintile workers gain almost nothing. The AI encodes tacit practices of the best agents and transfers them to novices. The within-firm skill distribution compresses. The staggered rollout provides within-firm variation. The authors do not address whether early adopters differ from late adopters in ways that bias the estimates. A firm that volunteers for an AI trial may differ from one that does not, and the agents who choose to use the tool may differ from those who do not. Selection of this kind could inflate the estimated gains, though the direction of the bias is not obvious.

The exposure literature broadens the picture beyond individual experiments. Large language models reach at least 10% of the tasks performed by roughly 80% of U.S. workers, and half or more of the tasks for about 19% (Eloundou et al., 2024). With complementary software, the technology could reach 47 to 56% of all tasks. Felten, Raj, and Seamans (2023) use a different methodology and find that exposure concentrates in higher-wage, higher-education occupations. The pattern reverses that of earlier automation. These exposure numbers are upper bounds, not predictions. They measure what is technically feasible. They do not measure what firms will actually do. That gap runs through the literature. It is why micro estimates of AI’s potential and macro estimates of its effects diverge.

Physical automation offers a cautionary precedent. Between 1990 and 2007, one additional industrial robot per thousand workers reduced local employment by 0.2% and wages by 0.42% in U.S. commuting zones. Displacement dominated reinstatement (Acemoglu & Restrepo, 2020). Whether AI follows this substitution pattern is an empirical question the literature has not resolved. Autor (2024) argues that AI could rebuild middle-class expertise in healthcare and software, if deployed to support decision-making rather than to replace workers. Rebuilding expertise requires institutional choices that steer deployment toward complementarity. The two perspectives pull in opposite directions. Autor sees AI as a tool for expanding middle-class work. Acemoglu and Restrepo document a history in which automation displaced workers and depressed wages. Both arguments rest on internally consistent logic. But neither has been tested against direct evidence on AI adoption at scale, over a horizon long enough to separate augmentation from displacement.

The experiments answer one question: AI raises productivity in the tasks studied. Whether these gains persist beyond experimental settings and produce aggregate effects? These are questions the macro literature takes up.

1.2 Macroeconomic adjustment: growth, output, and inflation

Macro estimates of AI’s effect on total factor productivity differ by an order of magnitude. The disagreement is less about model structure than about calibration. It reflects genuine

uncertainty about adoption.

The cautious benchmark comes from Acemoglu (2024). Applying Hulten’s theorem to a task-based framework, he estimates that AI will raise TFP by 0.53 to 0.66% over ten years. That amounts to roughly 0.06% per year. The logic is arithmetic. Exposure covers roughly 20% of tasks (Eloundou et al., 2024). Of those, 23% can be profitably automated within a decade. Per-task cost savings average 27%. The estimate rests on a specific assumption. Hulten’s theorem requires that marginal cost shares equal output elasticities. The condition holds under competition. It may break down if AI introduces non-marginal technology shifts, or if market power in AI provision is substantial. Hulten’s theorem is a first-order approximation. It captures the direct effect of a productivity improvement in one task on aggregate output, holding everything else constant. Whether AI induces the kind of structural change that the theorem cannot capture is what separates the macro optimists from the pessimists.

The same decomposition, applied with broader exposure and higher per-task gains, yields far larger numbers. Aghion and Bunel (2024) use 60% task exposure rather than 20%, and per-task gains averaging 40%. Their baseline is 0.68% of additional annual TFP growth. Their range spans 0.07 to 1.24. The 20% exposure figure comes from conservative task-level rubrics that exclude complementary software. The 60% figure includes it. Which number is right depends on whether firms build applications on top of foundation models. Exposure measures alone cannot answer that question. The two estimates capture different mechanisms. Acemoglu measures the direct substitution effect, while Aghion and Bunel include the indirect effect of complementary innovation. This is not a statistical disagreement. It turns on a bet about whether firms will make the organizational investments that allow AI’s effects to compound. Both papers use defensible assumptions, but neither provides evidence that its assumptions are the right ones.

General equilibrium forces can amplify or dampen the translation from task productivity to aggregate output. A multi-sector model by Aldasoro, Doerr, Gambacorta, and Rees (2024) treats AI as a sector-specific productivity shock calibrated to an industry-level exposure index. AI could raise annual productivity growth by about 1%. Inflation effects depend on expectations. Unanticipated gains are disinflationary. Anticipated gains trigger consumption and investment surges that generate inflationary pressure. The model also reveals that a sector’s initial AI exposure has little correlation with its long-run output gain. Input-output linkages redistribute productivity effects across the economy. This complicates a simple reading of exposure scores. The sectors that look most vulnerable today are not necessarily the ones that gain the most tomorrow. The model is calibrated than estimated. It tells us what could happen under a given set of assumptions, yet it cannot identify which set of assumptions is most likely.

The distributional effects extend across borders. Evidence from 56 economies, using a Rajan-Zingales identification strategy, suggests that generative AI benefits advanced economies more than emerging markets in the short run (Gambacorta, Kharroubi, Mehrotra, & Oliviero, 2026). The gap reflects sectoral composition and digital-infrastructure deficits. AI adoption may widen rather than narrow global income gaps. The identification strategy uses cross-industry variation in AI exposure interacted with country-level adoption trends. A standard concern with this approach is that industry-level exposure may correlate with other shocks that affect growth independently of AI. The authors control for broad sectoral

trends, but they cannot rule out the possibility that something other than AI is driving the differential growth patterns.

Beyond cross-country heterogeneity, a deeper uncertainty concerns whether AI changes the long-run growth rate or merely reallocates productivity gains across sectors. Baumol’s cost disease may constrain AI’s long-run growth effect. Aghion, Jones, and Jones (2018) model AI as task automation within a CES framework. Automated sectors become cheap and lose GDP weight. Bottleneck sectors determine the long-run growth rate. A growth explosion requires full automation of goods production, or automation of idea production itself. Current AI systems accomplish neither. Measured productivity has not yet responded. Brynjolfsson, Rock, and Syverson (2018) point to implementation lags. AI requires complementary intangible investment that national accounts do not capture. Organizational redesign takes time. The J-curve formalized by Brynjolfsson, Rock, and Syverson (2021) captures this dynamic. Early in adoption, resources devoted to intangibles depress measured productivity. Later, the accumulated stock generates output without being credited as input. By 2017, their estimates suggest true TFP exceeded official measures by 15.9%. The J-curve helps explain why economists can look at the same technology and reach TFP estimates that differ by a factor of ten. It does not, however, tell us where we are on the curve. If adoption is still early, measured productivity may understate true gains by a wide margin. If adoption has already plateaued, the J-curve explanation loses force. Distinguishing between these scenarios requires data on intangible investment that firms do not currently report.

1.3 Policy challenges and open questions

A recurring tension in the policy literature runs between distribution and efficiency. AI can raise aggregate welfare. It can also worsen inequality. Whether policy can preserve the gains without worsening the distribution is unresolved. The literature divides roughly into short-run and long-run questions. An environmental externality cuts across both.

In the short run, the question is who loses and how to compensate them. Korinek (2024) argues that AI threatens cognitive and creative occupations, not only routine work. Workers with specialized human capital may face larger income losses than the blue-collar workers displaced by earlier automation. Korinek’s argument is theoretical. It identifies which occupations are exposed. It does not estimate how many workers will actually be displaced, or over what horizon. That gap between exposure and displacement is the same measurement problem that runs through the micro and macro literatures. If the gains accrue mainly to capital owners and the firms that control the technology, the shift in factor shares erodes tax bases built on labor income. That erosion comes at a moment when fiscal capacity is already stretched. Cross-country evidence sharpens the concern. Gambacorta et al. (2026) show that the same digital-infrastructure deficits that limit AI adoption in emerging markets also limit the fiscal tools available to compensate displaced workers. Countries that gain the least from AI also have the weakest safety nets. AI may widen inequality both within and across countries. The countries least able to respond may face the largest adjustment costs.

The long-run question concerns tax design. When labor’s share of income declines permanently, the fiscal system must adapt. Thuemmel (2023) studies robot-specific taxes in a general equilibrium model with occupational choice. Optimal robot taxes are small, on the order of 1%. At current automation prices, a modest subsidy is welfare-improving.

It compresses the wage distribution from above. As robots become cheaper, the optimal policy switches to a tax. Thuemmel finds that the main welfare gains come from reforming the broader system of income and capital taxation. Robot-specific taxes capture a narrow slice of the distributional problem. The broader fiscal question is how to restructure income and capital taxation when labor’s share declines. Thuemmel’s model abstracts from several features that matter in practice. It assumes a static occupational structure. It does not model the creation of new tasks or occupations, which could offset some of the displacement. It treats robots and AI as equivalent, though AI may substitute for different types of labor than physical robots do. These abstractions are standard in the literature. They mean that the 1% estimate is best read as an order of magnitude, not a precise policy target.

An environmental externality cuts across both horizons. Bonfiglioli, Crinò, Filomena, and Gancia (2025) use a shift-share strategy across 722 U.S. commuting zones. Localities with higher AI penetration experienced slower emissions declines between 2002 and 2022. Without AI-driven growth, average emissions would have fallen approximately 37% more. The shift-share instrument uses pre-period industry composition interacted with national AI adoption trends. The result survives controls for local manufacturing decline. The authors do not report pre-trend tests across exposure groups, which would strengthen the case that the results reflect AI rather than pre-existing differences across localities. The mechanism is straightforward. Data centers require stable baseload power. Nearby plants shift generation toward fossil fuels to meet that demand. Training emissions for frontier models are growing rapidly (Stanford Institute for Human-Centered Artificial Intelligence, 2025). The externality is structural. Markets are unlikely to price it away without policy intervention. The environmental literature is younger than the productivity literature. The Bonfiglioli et al. study is among the first to link AI adoption to emissions at a spatially disaggregated level. More work is needed before the magnitude of AI’s carbon footprint can be estimated with confidence.

These debates turn on data that do not yet exist. Firm-level data linking AI adoption to productivity and intangible investment would let researchers test the parameter assumptions that now drive the order-of-magnitude disagreement in macro estimates. Without it, researchers cannot distinguish between a scenario where AI adds 0.06% to annual TFP growth and one where it adds 0.68. Worker-level panels that track displacement over longer horizons could separate short-run augmentation from long-run substitution. These data do not yet exist at the necessary scale. On the theoretical side, current models treat adoption costs as exogenous. Endogenizing them would let the theory explain why the same technology diffuses at different rates, rather than treating diffusion speed as a calibration choice. Combining data-center location with electricity-source variation would measure AI’s carbon footprint more precisely (Bonfiglioli et al., 2025; Brynjolfsson, Korinek, & Agrawal, 2025).

1.4 Concluding remarks

Generative AI raises productivity in cognitive tasks. The evidence from randomized experiments and workplace data agrees on this point. The largest gains go to workers at the bottom of the skill distribution. This finding separates AI from earlier automation technologies, which displaced routine physical and clerical labor and widened skill premiums from below.

Whether these micro gains aggregate into faster macroeconomic growth is unsettled. TFP

estimates differ by an order of magnitude. They rest on assumptions about adoption breadth and complementary investment that existing data cannot verify. The productivity-paradox literature documents a pattern. Intangible investment depresses measured productivity during adoption. The gains arrive later. Whether AI raises long-run growth rates by automating the production of new ideas is a further question. Growth theory takes this possibility seriously. No empirical test has addressed it.

The literature traces a pattern. The micro evidence is consistent across experimental and workplace settings. That degree of convergence is unusual this early in a technology’s diffusion. The macro implications remain unsettled. The gap between the two is a reflection of genuine uncertainty. How does a technology that raises productivity in specific tasks translate into economy-wide outcomes? The next step is empirical. Observing adoption directly, rather than inferring it from occupational exposure scores, requires data that no one has yet collected.

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